

Topology Is Not a Particle Label: A Typed-State Requirement for a Filament Particle Taxonomy

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SAT/H(s)H Sandbox Geometry Program
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A historical SAT Particle Zoo source assigns particle flavor to resonance mode, places multiple flavors in common filament classes, represents force carriers as ripples or transitions, and represents baryons as braided composites. These definitions imply a simple but consequential formal restriction: particle identity cannot be represented faithfully by carrier topology alone. We state a minimal typed-state decomposition and demonstrate the non-injectivity geometrically with an explicit harmonic filament family whose carrier topology is unchanged while curvature, torsion, and arclength vary. A separate exact three-strand representative demonstrates that the stated braid geometry can be generated reproducibly while also exposing the need for endpoint, framing, and binding conventions before assigning a particle-specific topological invariant. The result is a source-faithful taxonomy constraint, not a claim that the historical particle identifications are physically realized.

I. SOURCE-DERIVED TYPING PROBLEM

The controlling historical source for this note is `SATv THE PARTICLE ZOO.txt`. It describes fermions as persistent filaments, particle flavor as a resonance mode, quarks as short high-curvature filament segments that bind into braided composites, leptons as longer singly curved filaments, neutrinos as near-aligned filaments, and bosons as ripples or structural transitions.

These are not all the same mathematical object type. A minimal state is therefore

$$X = (C, F, G, E, B, P), \quad (1)$$

where C is carrier class, F is framing/topological data, G is metric or differential geometry, E is excitation/phase state, B is binding/composite relation, and P is persistence/event type.

Let $\pi_C : X \rightarrow C$ retain only the carrier class. Because the source distinguishes flavors through resonance/excitation or geometry while retaining the same broad carrier class, the particle-labeling map is not injectively recoverable from π_C alone.

II. CALCULATED SAME-TOPOLOGY FAMILY

Consider the family of embedded filament segments

$$\gamma_n(s) = (R \cos ns, R \sin ns, ps), \quad 0 \leq s \leq 4\pi, \quad (2)$$

for positive integers n . Every member has interval carrier topology, while

$$\kappa_n = \frac{Rn^2}{R^2n^2 + p^2}, \quad (3)$$

$$\tau_n = \frac{pn}{R^2n^2 + p^2}. \quad (4)$$

Thus geometry and harmonic state can vary while carrier topology is unchanged.

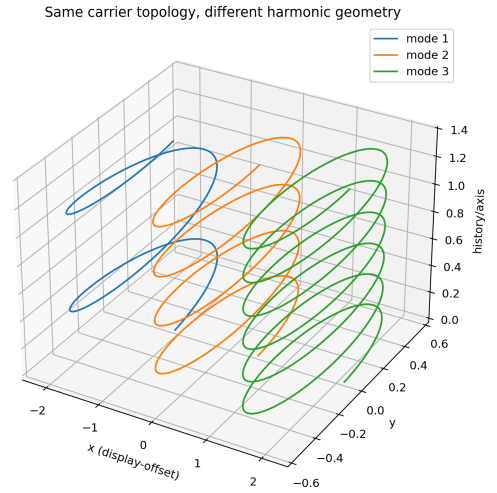


FIG. 1. Three exact members of Eq. (2), displayed with lateral offsets only for legibility. The construction is a representative realization of the historical source’s “harmonic” distinction, not a fitted electron–muon–tau model.

III. BRAIDED COMPOSITE REPRESENTATIVE

The source describes a baryonic composite as three braided quark filaments. A minimal exact representative is

$$\begin{aligned} \beta_j(z) = & (R_b \cos(2\pi z + 2\pi j/3), \\ & R_b \sin(2\pi z + 2\pi j/3), z), \end{aligned} \quad (5)$$

$j = 0, 1, 2,$

for $0 \leq z \leq 1$. At equal z , every strand pair is separated by

$$d = \sqrt{3} R_b, \quad (6)$$

which supplies a direct numerical regression check.

For open strands, the topology depends on what is held fixed at the boundary and on whether framing is

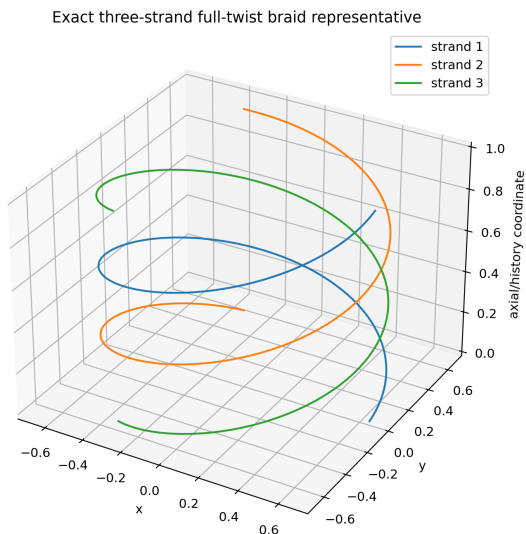


FIG. 2. Exact three-strand full-twist representative from Eq. (5). It realizes the stated three-filament braid geometry but is not, by itself, a proton or neutron identification.

part of the state. A centerline alone also does not record ribbon or worldtube twist. Particle-specific topological

assignments therefore require endpoint, framing, binding, and readout conventions beyond the carrier curves in Eq. (5).

IV. RELATED USE CASES

Topological particle taxonomies using braids or knots have precedents, including Bilson-Thompson, Hackett and Kauffman’s braided-belt construction, Finkelstein’s quantum-knot program, and Asselmeyer-Maluga’s braid/3-manifold constructions. These are comparison points for mathematical use-case only; no equivalence to SAT/H(s)H is asserted.

V. CONSEQUENCE FOR SOLVER DESIGN

A Particle Zoo solver should return a typed state rather than a single topology label. In particular, the implementation should not force a unique knot or braid invariant onto a particle whose source definition distinguishes it by resonance, curvature, phase, or composite binding. This separation permits topological invariants to be used where defined while retaining metric and dynamical information required by the source taxonomy.

[1] S. O. Bilson-Thompson, J. Hackett, and L. H. Kauffman, “Particle Topology, Braids, and Braided Belts,” *J. Math. Phys.* **50** (2009), doi:10.1063/1.3237148.

[2] R. J. Finkelstein, “The Elementary Particles as Quantum Knots in Electroweak Theory,” *Int. J. Mod. Phys. A* **22** (2007), doi:10.1142/S0217751X0703707X.

[3] T. Asselmeyer-Maluga, “Braids, 3-Manifolds, Elementary Particles,” *Symmetry* **11** (2019), doi:10.3390/sym11101298.